StreetRace Manager
Integration Testing Report

March 23, 2026

1 Introduction

This report documents the design, integration, and testing of the StreetRace Manager system. The system manages underground street races, crew members, vehicles, inventory, results, and missions. Each module was implemented separately and then integrated to verify correct interaction and data flow across modules.

2 Modules Implemented

- Registration
- Crew Management
- Inventory
- Race Management
- Results
- Mission Planning
- Vehicle Maintenance (additional)
- Crew Availability (additional)

3 Additional Modules

Vehicle Maintenance: Inspects and repairs vehicles. It checks for available mechanics, consumes parts and tools, and updates car condition.

Crew Availability: Tracks whether crew members are currently free or busy. Used by race assignment and mission planning to enforce availability constraints.

4 Business Rules Implemented

- A crew member must be registered before a role can be assigned.
- Only crew members with the driver role may enter races.
- Drivers must be available before being assigned to races.
- Cars must exist and be in usable condition (not damaged) to be assigned to races.
- Race results update cash balance and car condition.
- Driver availability is restored after race results are recorded.
- All required roles must be covered before a mission can start.
- Assigned crew must be available before mission start.
- Mission start marks crew unavailable; completion marks them available again.
- Vehicle repair requires a mechanic and consumes parts/tools.

5 Call Graph

A hand-drawn call graph was created showing all functions and their call relationships within and between modules. Each node represents a single function and arrows indicate function calls, with cross-module interactions clearly marked.

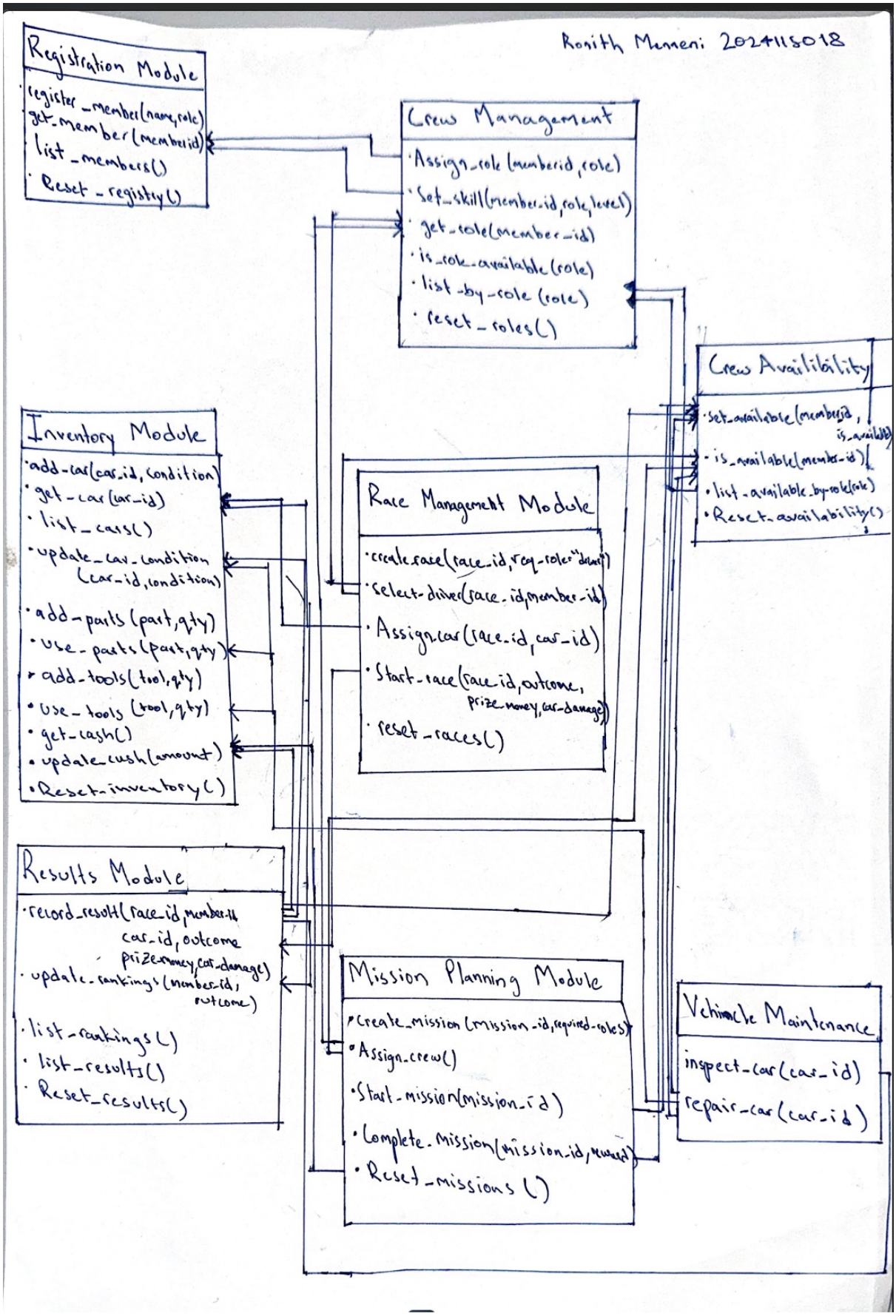

Figure 1: Enter Caption

6 Integration Test Design and Results

Integration tests validate interactions between modules and ensure that business rules are enforced across module boundaries.

Scenario	**Modules**	**Expected Result**	**Actual Result**	**Notes**
Register driver then enter race	Registration, Crew Management, Inventory, Race Management	Driver accepted and car assigned	driver_ok=True, car_ok=True	Validates role check and car lookup
Enter race without driver	Registration, Crew Management, Race Management	Driver rejected because role is not driver	driver_ok=False	Role validation works
Race results update inventory	Race Management, Results, Inventory	Cash increases and car condition becomes damaged	cash=500, car_condition=damaged	Results pipeline correct
Mission assignment and start	Mission Planning, Crew Management, Crew Availability	Mission assigns and start sets crew unavailable	assigned_ok=True, started_ok=True, driver_available=False, mechanic_available=False	Availability transitions correct
Vehicle repair with mechanic	Vehicle Maintenance, Inventory, Crew Availability, Crew Management	Repair succeeds and car condition becomes ok	repaired_ok=True, car_condition=ok	Repair rules enforced
Unavailable driver cannot enter race	Crew Availability, Race Management, Crew Management	Driver rejected because not available	driver_ok=False	Availability enforced
Damaged car cannot be assigned	Inventory, Race Management	Car rejected because condition is damaged	car_ok=False	Car condition enforced
Mission completion restores availability	Mission Planning, Crew Availability, Inventory	Driver available again and cash increased	driver_available=True, cash=150	Completion updates availability and cash

7 Conclusion

All integration tests passed and the expected outcomes matched the actual results. The modules interact correctly and the required business rules are enforced. The two additional modules (Vehicle Maintenance and Crew Availability) add realistic constraints and are fully integrated into race and mission workflows.